



Government College University Faisalabad
Government Graduate College Satiana Road, Faisalabad
Department of Computer Science
(PROJECT PROPOSAL)



DATE

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Project Code :

Project Title : AgriPulse Soil-Test Driven Farm Planner that Tells a Punjab Farmer His Cost, His Water Need, His Fertilizer Bags and His Profit, Before He Sows.

Problem Statement:

A farmer in Punjab must decide what to sow, how much to spend, and how much fertiliser to buy before the season starts. Almost every tool that exists today asks him for answers he does not have, or needs a signal his village does not get.

Pakistani farming decisions are made under three conditions at once: the farmer has very little money, the decision is made once and cannot be undone until harvest, and the information needed to make the decision is locked inside government offices and paid services.

We studied 13 existing tools 5 working in Pakistan (Bakhabar Kissan, Kissan, Agrixia, MandiOye, PAR) and 8 international ones (PlantVillage/Nuru, Leafwise, LeafAid AI, Kisan Sarathi, Kisan Suvidha, Miraitu, KisanPe, CropKhata/Farm Manager).

We found four separate problems.

- Farming decisions are made without real information. The soil test already exists as a card of numbers, but nobody turns it into a crop choice, a fertiliser quantity or a cost sheet so the crop is chosen blind and money and water are wasted on the wrong dose.
- There is no planning before the harvest. On the day of sowing the farmer does not know his total expense, his water requirement, his expected yield or his profit; he plans in his head from last year's prices, cannot compare two crops side by side, and finds out the cost only when the season ends.
- Disease names and cures are in a foreign language. Advisory books are written in English while the farmer reads Urdu, there is no photo to tell two diseases apart, and no remedy is given in his own language. This problem is not solved in this project it is future work.
- Existing tools depend on hardware and paid services. They need physical weather stations, a working internet connection, a monthly data subscription, or a trained model on a server so in the field the farmer either pays or goes blind.

Tools to be used:

- =>React 18
- =>Vite
- =>Tailwind CSS
- =>Recharts
- =>React Router
- =>Fetch API client
- =>Node.js
- =>Express.js
- =>MongoDB
- =>MongoDB Compass
- =>JWT
- =>Bcrypt
- =>Zod
- =>Cors
- =>Dotenv
- =>Morgan
- =>Git + GitHub

Aim and Objective:

To build AgriPulse, an web application that turns a soil test card and a land size into the four decisions a farmer must make before sowing cost, water, fertilizer bags and profit in Pakistani Rupees and liters, with every number showing the government or FAO publication it came from, so that the farmer can decide without paying anyone, without an internet connection, and without a subscription.

Each objective is written so that it can be marked pass or fail.

- O1.** The system shall read a soil test card and show the farmer what his soil is short of, using a radar chart that plots his value against the target range for his soil type.
- O2.** The system shall calculate the exact number of Urea, DAP and MOP bags using the Government of the Punjab SFRI Guide-V formulator an invented formula and the result must land on 2.35 bags Urea and 1.04 bags DAP per acre for cotton, the figure published in the PCPA Cotton Book.
- O3.** The system shall calculate the total cost of sowing, broken down line by line into land preparation, seed, fertilizer, labor, water and transport, with every line matching the PCPA Cotton Book or the PAU Package of Practices figure for that crop.
- O4.** The system shall calculate the total water need in liters and the number of irrigations using the FAO-56 method, with crop coefficients cited from FAO-56 Table 12.
- O5.** The system shall show the expected yield, net profit, ROI and the break-even price, using the PAU district yield band for yield and NFDC for price.
- O6.** The system shall let the farmer move sliders and see cost, profit and ROI change immediately, so that he can compare crops before committing, with the recompute happening in the browser and with zero network calls.
- O7.** The system shall work fully offline, so that field signal is never a problem every screen still opens and every calculation still runs with the router off.
- O8.** The system shall show the source of every number on the screen, next to the figure rather than hidden behind a help icon, so that both the farmer and the supervisor can trust it.
- O9.** The system shall keep the farmer's data private one farmer can never see an other's using JWT login where every protected query is filtered by the logged-in user's ID.

Scope of the Project:

The project delivers a five-page offline web application built from four functional modules, served by six API route files and nine backend packages. The farmer registers once, records his district, land area and soil type, enters the N-P-K numbers from his soil test card, and receives the exact fertiliser bags, the full cost breakdown, the water requirement in litres, the expected yield, the net profit, the ROI and the break-even price each figure carrying the name of its public source. A what-if simulator then lets him move sliders on area, budget and crop and watch cost, profit and ROI change live, so that two crops can be compared side by side before money is committed. The entire application runs on one laptop with the internet switched off, and it holds no machine-learning model of any kind.

What the project will build:

- M0 Login and Register. Email and password with JWT, one account per farmer. (Page 1)
- M1 Land and Climate Profiler. District, area in acres, soil type, and the climate card for that district and month. (Pages 2 and 3)
- M2 Resource and Financial Planner. Full cost breakdown, water in litres, yield, net profit, ROI and

break-even price. (Page 4A)

- M3 NPK Soil Fertiliser Advisor. Deficiency ranking, exact Urea, DAP and MOP bag counts, fertiliser cost, and the radar chart. (Page 5)
- M4 What-If Simulator. Sliders for area, budget and crop, live recompute, and reset to baseline. (Page 4B)

Coverage and data volume:

- 5 crops wheat, rice, maize, potato, tomato (the five with both agronomy and disease data available for Punjab).
- 5 soil types sandy, sandy loam, loam, clay loam, clay.
- 30 districts of Punjab and Sindh, from the provincial Agriculture Departments, with 360 climate records (30 districts × 12 months).
- 420 total reference rows, loaded by one seed script.
- 3 user collections user, land, plan. 0 model files in the repository.

Functional Requirements	Non-Functional Requirements
FR-A1 A visitor can register with full name, email and password.	NFR-1 Offline The app works with the network completely off, including after a page refresh.
FR-A2 A registered farmer can log in and receives a JWT expiring after 24 hours.	NFR-2 Offline No runtime call to any external API, CDN, font host or analytics service.
FR-A3 Passwords are never stored in plain text.	NFR-3 Performance A page appears in under 2 seconds on a mid-range laptop; Lighthouse performance ≥ 90.
FR-A4 A duplicate email is rejected with 409.	NFR-4 Performance Any single calculation finishes in under 300 ms.
FR-A5 A protected page cannot be opened without a valid token.	NFR-5 Performance Slider recompute finishes in under 50 ms with no visible lag.
FR-A6 A farmer cannot read another farmer's land or plan.	NFR-6 Security Passwords are stored only as bcrypt hashes, never in plain text.
FR-B1 The farmer selects a district from a list of 30.	NFR-7 Security Every protected endpoint requires a valid JWT.
FR-B2 The farmer enters land area in acres, from 0.1 to 100.	NFR-8 Security No user can read or modify another user's data.
FR-B3 The farmer selects a soil type from 5 options.	NFR-9 Security All secrets come from environment variables, never from source code.
FR-B4 The system shows the climate card for the selected district and month.	NFR-10 Data integrity Every calculated number is traceable to a named public source; each API response carries a source field.
FR-B5 The climate card shows irrigation need and frost risk.	NFR-11 Correctness The four calculation services are covered by unit tests; npm test is green with at least 25 test cases.
FR-B6 Land details are saved against the logged-in farmer.	
FR-C1 The farmer enters nitrogen, phosphorus and potassium in kg/ha from his soil test card.	

FR-C2 The system converts ppm to kg/ha for the farmer's area. FR-C3 The system ranks each nutrient as Poor, Medium or Adequate using the SFRI band table for that soil type. FR-C4 The system subtracts the soil value from the crop requirement to get the nutrient needed. FR-C5 The system converts the nutrient needed into whole bags of Urea, DAP and MOP, rounding up. FR-C6 The system multiplies per-acre bags by acres to get total bags. FR-C7 The system shows a radar chart of soil health against the target ring for that soil type. FR-C8 The system shows 0 bags when the nutrient is already above target. FR-C9 The system shows a "How this was calculated" panel on screen, naming SFRI Punjab Guide-V. FR-C10 The system shows the fertiliser cost in PKR as bags × NFDC retail price. FR-C11 (validation test) For cotton the calculator must return 2.35 bags Urea and 1.04 bags DAP per acre, matching the PCPA Cotton Book. FR-D1 The farmer selects one of 5 crops. FR-D2 The farmer selects the sowing month, from 1 to 12. FR-D3 The farmer enters an optional budget in PKR, which produces a headroom warning. FR-D4 The system returns a cost breakdown with named lines land preparation, seed, urea, DAP, labour, water, transport each with its PKR amount. FR-D5 Each cost line shows its source on screen. FR-D6 The system calculates total water in litres and the number of irrigations using FAO-56 ($ET_c = \Sigma(ET_o \times K_c \times \text{days})$). FR-D7 The system calculates expected yield in maund using the district yield band. FR-D8 The system calculates net profit and ROI. FR-D9 The system calculates the break-even price per maund. FR-D10 The system warns in red when cost exceeds budget.	NFR-12 Reliability Out-of-range months and non-numeric areas return 400, not 500. NFR-13 Usability The flow is completed in under 3 minutes by someone who has never seen the app. NFR-14 Usability No screen requires horizontal scrolling on a 360px phone. NFR-15 Accessibility Interactive controls are keyboard reachable with a visible focus ring. NFR-16 Accessibility Colour is never the only signal; deficiency also shows the word LOW and excess shows HIGH. NFR-17 Maintainability No backend file exceeds 250 lines; business logic lives in services, never in routes. NFR-18 Portability The app runs on Windows, Linux and macOS with one command from a fresh clone. NFR-19 Privacy No farmer data leaves the machine; there is no telemetry of any kind. NFR-20 Documentation Sources are visible on screen, not behind a help icon.
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FR-D11 The system shows a cost bar chart and three stat cards for water, yield and profit. FR-E1 The farmer can move a slider for area, budget and crop, each keyboard accessible with 44px touch targets. FR-E2 Numbers update as the slider moves, with no button press. FR-E3 The simulator makes no network calls while sliders move. FR-E4 The simulator writes nothing to the database. FR-E5 The screen shows baseline and scenario side by side with the change as a percentage. FR-E6 A "Reset to baseline" button restores the original plan. FR-E7 The screen states plainly that results are session-only. FR-F1 One seed script loads all reference data into MongoDB, reporting 420 reference rows inserted. FR-F2 The script is safe to run twice — duplicates are skipped, no crash. FR-F3 The API serves reference lists without a login.	
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Particulars of the students:

Sr. #	Registration	Name in Full	Email	Contact #	CGPA	Signatures
1						
2						

Remarks:	Signatures and Date
Approved: ☐ YES ☐ NO	Coordinator Head Of Department